

WELKOM, South Africa

WMO: 683450

Lat: 27.995S

Lon: 26.666E

Elev: 4409

StdP: 12.50

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.3	35.0	9.0	10.3	60.9	12.3	12.1	60.9	22.6	61.7	20.5	59.7	5.9	120	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	22.2	93.9	61.0	91.6	61.0	89.4	60.8	69.1	81.1	68.0	79.9	67.0	78.8	10.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	112.9	72.5	64.7	107.9	71.7	63.6	103.9	71.4	36.1	81.2	35.1	79.7	34.3	78.9	82.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 22.4	19.9	17.7	DB	28.1	98.4	2.5	3.3	26.3	100.7	24.8	102.7	23.3	104.5	21.5	106.9	(4)
(5)			WB	24.7	71.0	2.3	4.1	23.0	73.9	21.7	76.4	20.4	78.7	18.8	81.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	74.8	74.2	70.8	64.6	58.2	52.6	52.1	57.1	64.5	68.7	71.5	73.7	(6)
(7)		DBStd	9.56	4.22	3.85	3.94	4.80	5.23	4.77	4.89	6.86	6.45	5.82	5.84	4.49	(7)
(8)		HDD50	92	0	0	0	0	8	28	35	20	1	0	0	0	(8)
(9)		HDD65	1447	1	1	4	62	216	372	400	255	86	31	16	3	(9)
(10)		CDD50	5635	769	677	645	439	261	106	100	242	437	580	645	734	(10)
(11)		CDD65	1515	306	258	184	52	4	0	0	11	71	146	212	271	(11)
(12)		CDH74	15099	2730	2197	1531	523	127	7	4	220	1045	1741	2358	2616	(12)
(13)		CDH80	5783	1161	868	478	73	11	0	0	32	331	675	1032	1122	(13)
(14)	Wind	WSAvg	8.4	8.9	8.3	7.9	6.8	6.2	6.7	7.0	8.7	9.4	9.9	10.3	10.1	(14)
(15)	Precipitation	PrecAvg	19.7	3.0	2.2	3.1	1.5	0.9	0.3	0.2	0.4	0.5	1.9	2.7	3.3	(15)
(16)		PrecMax	30.6	6.7	6.4	5.5	5.4	6.1	1.2	1.7	2.2	2.3	5.4	7.0	6.1	(16)
(17)		PrecMin	11.2	0.9	0.5	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.1	0.8	(17)
(18)		PrecStd	5.9	1.5	1.3	1.7	1.4	1.4	0.3	0.4	0.7	0.5	1.4	1.9	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.3	94.2	93.0	85.7	81.3	74.5	74.1	83.6	89.9	93.4	95.5	96.9	(19)
(20)			MCWB	62.2	61.4	63.3	59.3	55.3	50.7	49.9	54.8	55.9	57.5	60.7	60.5	(20)
(21)		2%	DB	92.9	91.2	88.2	82.0	77.3	71.0	71.2	79.5	86.8	90.3	92.3	92.7	(21)
(22)			MCWB	63.2	63.0	61.9	59.9	53.7	49.5	49.2	52.6	55.3	57.7	59.5	61.2	(22)
(23)		5%	DB	89.6	88.3	85.2	79.7	74.7	68.4	68.4	76.1	84.0	87.4	89.5	89.5	(23)
(24)			MCWB	63.0	64.3	62.3	59.4	52.9	48.8	48.5	51.6	54.2	57.1	59.6	61.4	(24)
(25)		10%	DB	86.6	85.6	82.5	77.1	71.8	66.0	65.7	72.5	80.6	84.0	86.6	86.4	(25)
(26)			MCWB	63.7	64.9	62.5	58.0	52.4	48.1	47.5	50.1	53.4	56.8	59.4	61.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.1	70.5	69.9	65.4	59.1	54.8	53.8	57.2	60.1	64.1	67.0	69.1	(27)
(28)			MCDWB	82.3	82.7	79.9	76.5	72.4	66.7	66.4	76.4	76.8	78.7	81.8	81.7	(28)
(29)		2%	WB	68.5	69.3	67.8	63.5	56.9	52.6	51.3	54.8	58.1	62.6	65.3	67.5	(29)
(30)			MCDWB	80.5	81.5	78.7	74.8	70.6	65.7	66.5	75.1	77.2	77.9	80.2	80.0	(30)
(31)		5%	WB	67.5	68.1	66.3	62.1	55.4	50.9	49.8	53.2	56.6	61.2	63.9	66.3	(31)
(32)			MCDWB	79.3	80.2	77.6	73.2	69.1	64.0	65.8	71.9	76.9	76.3	79.0	78.7	(32)
(33)		10%	WB	66.6	67.0	64.8	60.8	53.9	49.2	48.1	51.5	55.2	59.9	62.8	65.1	(33)
(34)			MCDWB	78.0	78.6	76.4	72.6	67.9	62.4	62.8	68.5	75.6	74.7	77.9	77.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	22.2	21.2	22.3	23.0	27.0	26.9	26.5	27.9	28.9	26.7	26.2	23.9	(35)
(36)			MCDBR	27.2	26.3	26.7	26.3	30.0	29.6	29.3	30.6	32.0	31.0	29.9	29.0	(36)
(37)			MCWBR	7.4	7.7	8.7	9.6	12.0	13.6	13.4	13.1	12.7	10.8	8.9	8.4	(37)
(38)		5% WB	MCDWB	20.7	20.2	21.3	21.2	24.8	25.3	25.9	27.6	27.9	24.1	23.9	21.7	(38)
(39)			MCWBR	7.6	7.4	8.8	8.4	11.2	12.6	12.3	12.5	11.7	9.1	8.6	7.9	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.361	0.364	0.351	0.313	0.284	0.269	0.268	0.321	0.360	0.384	0.339	0.346	(40)
(41)		taud		2.375	2.371	2.381	2.467	2.523	2.532	2.514	2.350	2.237	2.212	2.385	2.404	(41)
(42)		Ebn at Noon		311	305	298	296	292	290	294	288	291	295	316	317	(42)
(43)		Edh at Noon		41	40	38	31	27	25	26	35	43	47	41	40	(43)
(44)	All-Sky Solar Radiation	RadAvg		2312	2103	1852	1537	1323	1190	1300	1564	1913	2135	2347	2396	(44)
(45)		RadStd		205	200	121	105	62	55	39	78	87	125	143	127	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+1.17	+1.37	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page